

STEP 1: Install Required Libraries

```
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.14)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.13)
Requirement already satisfied: thinc<8.4.0,>=8.3.12 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.13)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.3)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.0)
Requirement already satisfied: confection<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.3.3)
Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.1)
Requirement already satisfied: numpy<=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic<3.0.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.12.3)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging<=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types<=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0->sp)
Requirement already satisfied: pydantic-core<=2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0->sp)
Requirement already satisfied: typing-extensions<=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0)
Requirement already satisfied: typing-inspection<=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac)
Requirement already satisfied: certifi<=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.12->spacy) (1.3.0)
Requirement already satisfied: shellingham<1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (1.5.4)
Requirement already satisfied: rich<=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (13.9.4)
Requirement already satisfied: annotated-doc<=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (0.0.2)
Requirement already satisfied: cloudpathlib<=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->spacy) (0.19.0)
Requirement already satisfied: smart-open<=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->spacy) (5.2.1)
Requirement already satisfied: httpx<=0.24.0 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->spacy) (0.28.1)
Requirement already satisfied: MarkupSafe<=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->spacy) (3.0.3)
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Requirement already satisfied: httpcore<=1.* in /usr/local/lib/python3.12/dist-packages (from httpx<=0.24.0->weasel<2.0.0,>=1.0.0)
Requirement already satisfied: h11<=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore<=1.*->httpx<=0.24.0->weasel<2.0.0)
Requirement already satisfied: markdown-it-py<=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich<=12.3.0->typer<1.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich<=12.3.0->typer<1.0.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<=5.2.1->weasel<2.0.0,>=1.0.0->sp)
Requirement already satisfied: mdurl<=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py<=2.2.0->rich<=12.3.0->
Collecting en-core-web-sm==3.8.0
Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any
12.8/12.8 MB 91.2 MB/s eta 0:00:00
```

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

STEP 2: Import Libraries

```
import nltk
import spacy
from nltk.tokenize import sent_tokenize, word_tokenize
from nltk.stem import PorterStemmer
from nltk.corpus import stopwords
```

```
# Download required NLTK data
nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
```

```
[nltk_data] Unzipping corpora/stopwords.zip.
True
```

STEP 3: Load Medical Text Corpus

```
medical_text = """
Diabetes mellitus is a chronic condition characterized by high blood sugar levels.
Patients may experience symptoms like frequent urination, increased thirst, and fatigue.
Treatment includes insulin therapy, dietary changes, and regular physical activity.
Hypertension is another common disease that can lead to serious complications like heart attack and stroke.
"""
```

STEP 4: Sentence Tokenization (NLTK)

```
nltk.download('punkt_tab')
sentences = sent_tokenize(medical_text)

print("Sentences:")
for s in sentences:
    print(s)
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
Sentences:
```

```
Diabetes mellitus is a chronic condition characterized by high blood sugar levels.
Patients may experience symptoms like frequent urination, increased thirst, and fatigue.
Treatment includes insulin therapy, dietary changes, and regular physical activity.
Hypertension is another common disease that can lead to serious complications like heart attack and stroke.
```

STEP 5: Word Tokenization (NLTK)

```
words = word_tokenize(medical_text)

print("\nWords:")
print(words)
```

```
Words:
['Diabetes', 'mellitus', 'is', 'a', 'chronic', 'condition', 'characterized', 'by', 'high', 'blood', 'sugar', 'levels', '.', 'Pat
```

STEP 6: Tokenization using spaCy

```
nlp = spacy.load("en_core_web_sm")

doc = nlp(medical_text)

spacy_tokens = [token.text for token in doc]

print("\nspace Tokens:")
print(spacy_tokens)
```

```
space Tokens:
['\n', 'Diabetes', 'mellitus', 'is', 'a', 'chronic', 'condition', 'characterized', 'by', 'high', 'blood', 'sugar', 'levels', '.']
```

STEP 7: Stemming (NLTK)

```
ps = PorterStemmer()

stemmed_words = [ps.stem(word) for word in words if word.isalpha()]

print("\nStemmed Words:")
print(stemmed_words)
```

```
Stemmed Words:
['diabet', 'mellit', 'is', 'a', 'chronic', 'condit', 'character', 'by', 'high', 'blood', 'sugar', 'level', 'patient', 'may', 'e
```

STEP 8: Lemmatization (spaCy)

```

lemmatized_words = [token.lemma_ for token in doc if token.is_alpha]

print("\nLemmatized Words:")
print(lemmatized_words)

```

```

Lemmatized Words:
['diabetes', 'mellitus', 'be', 'a', 'chronic', 'condition', 'characterize', 'by', 'high', 'blood', 'sugar', 'level', 'patient',

```

STEP 9: Remove Stopwords

```

stop_words = set(stopwords.words('english'))

filtered_words = [word for word in words if word.lower() not in stop_words and word.isalpha()]

print("\nFiltered Words (No Stopwords):")
print(filtered_words)

```

```

Filtered Words (No Stopwords):
['Diabetes', 'mellitus', 'chronic', 'condition', 'characterized', 'high', 'blood', 'sugar', 'levels', 'Patients', 'may', 'experi

```

STEP 10: Comparison (Stemming vs Lemmatization)

```

print("\nComparison (First 15 words):\n")

for i in range(min(15, len(stemmed_words), len(lemmatized_words))):
    print(f"Original: {words[i]} | Stemmed: {stemmed_words[i]} | Lemmatized: {lemmatized_words[i]}")

```

```

Comparison (First 15 words):

Original: Diabetes | Stemmed: diabet | Lemmatized: diabetes
Original: mellitus | Stemmed: mellitu | Lemmatized: mellitus
Original: is | Stemmed: is | Lemmatized: be
Original: a | Stemmed: a | Lemmatized: a
Original: chronic | Stemmed: chronic | Lemmatized: chronic
Original: condition | Stemmed: condit | Lemmatized: condition
Original: characterized | Stemmed: character | Lemmatized: characterize
Original: by | Stemmed: by | Lemmatized: by
Original: high | Stemmed: high | Lemmatized: high
Original: blood | Stemmed: blood | Lemmatized: blood
Original: sugar | Stemmed: sugar | Lemmatized: sugar
Original: levels | Stemmed: level | Lemmatized: level
Original: . | Stemmed: patient | Lemmatized: patient
Original: Patients | Stemmed: may | Lemmatized: may
Original: may | Stemmed: experi | Lemmatized: experience

```